

Expelexia Lab - Data Analysis Report

File Name: data.csv
Date of Analysis: 2026-03-26 15:58
Prepared For: Expelexia (AI Analytics Program)
Project: Expelexia Lab

This report provides a comprehensive analysis and AI-generated insights based on the uploaded data.

Executive Summary

This report summarizes analysis results for data.csv using the same data shown in the website dashboard. The dataset contains 944 records across 10 variables. Main areas needing attention: footfall: 18 anomalies, high variance, tempMode: high variance, AQ: high variance. Observed trend signals: footfall down, tempMode up, AQ down. Recommendations focus on practical monitoring, early detection of unusual changes, and improving reliability of future data collection.

Data Overview

Measure	Value	What it means
Number of records	944	How many entries were analyzed.
Number of variables	10	How many fields were evaluated.
Missing values	{'footfall': 0, 'tempMode': 0, 'AQ': 0, 'USS': 0, 'CS': 0, 'VOC': 0, 'RP': 0, 'IP': 0, 'Temperature': 0, 'fail': 0}	Missing entries can reduce result reliability if frequent.

Website Dashboard Alignment

This table mirrors key trend, focus, and confidence indicators used in the website dashboard view.

Metric	Trend	Focus Area	Confidence
footfall	down	18 anomalies, high variance	0.980
tempMode	up	high variance	1.000
AQ	down	high variance	1.000
USS	up	high variance	1.000
CS	down	13 anomalies, high variance	0.990
VOC	up	high variance	1.000
RP	up	high variance	1.000
IP	down	high variance	1.000

Metric	Trend	Focus Area	Confidence
Temperature	up	high variance	1.000
fail	stable	high variance	1.000

Visual Analysis

The chart below shows how key values compare. It helps identify increases, decreases, and unusual patterns over time.

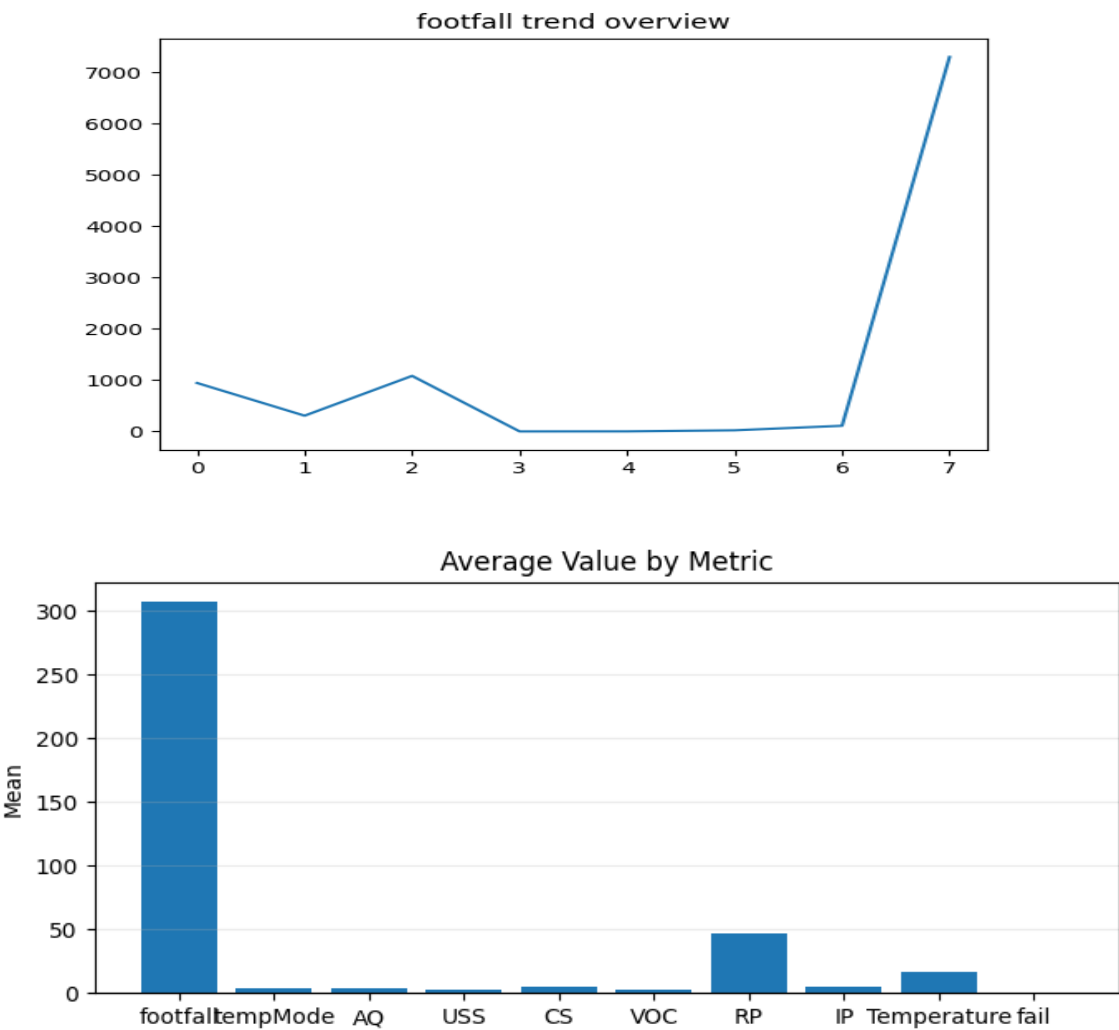


Chart Interpretation:

- footfall: bar chart centers near mean 306.381; line trend indicates down. Mean/median relation suggests a right-skewed distribution.
- tempMode: bar chart centers near mean 3.728; line trend indicates up. Mean/median relation suggests a right-skewed distribution.
- AQ: bar chart centers near mean 4.325; line trend indicates down. Mean/median relation suggests a right-skewed distribution.
- USS: bar chart centers near mean 2.940; line trend indicates up. Mean/median relation suggests a left-skewed distribution.
- CS: bar chart centers near mean 5.394; line trend indicates down. Mean/median relation suggests a left-skewed distribution.

- VOC: bar chart centers near mean 2.842; line trend indicates up. Mean/median relation suggests a right-skewed distribution.
- RP: bar chart centers near mean 47.043; line trend indicates up. Mean/median relation suggests a right-skewed distribution.
- IP: bar chart centers near mean 4.566; line trend indicates down. Mean/median relation suggests a right-skewed distribution.
- Temperature: bar chart centers near mean 16.332; line trend indicates up. Mean/median relation suggests a left-skewed distribution.
- fail: bar chart centers near mean 0.416; line trend indicates stable. Mean/median relation suggests a right-skewed distribution.
- Histogram bins show where value density is concentrated between quartile ranges.
- Pie chart summarizes stable vs attention-needed metrics from focus-area rules.

Data Change Explanation

Data Change Analysis:

- footfall: value range 0.000 to 7300.000 (spread 7300.000), middle 50% spans 109.000. Trend is down; volatility appears high; focus area is 18 anomalies, high variance.
- tempMode: value range 0.000 to 7.000 (spread 7.000), middle 50% spans 6.000. Trend is up; volatility appears high; focus area is high variance.
- AQ: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 3.000. Trend is down; volatility appears high; focus area is high variance.
- USS: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 2.000. Trend is up; volatility appears high; focus area is high variance.
- CS: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 1.000. Trend is down; volatility appears high; focus area is 13 anomalies, high variance.
- VOC: value range 0.000 to 6.000 (spread 6.000), middle 50% spans 4.000. Trend is up; volatility appears high; focus area is high variance.
- RP: value range 19.000 to 91.000 (spread 72.000), middle 50% spans 24.000. Trend is up; volatility appears high; focus area is high variance.
- IP: value range 1.000 to 7.000 (spread 6.000), middle 50% spans 3.000. Trend is down; volatility appears high; focus area is high variance.
- Temperature: value range 1.000 to 24.000 (spread 23.000), middle 50% spans 7.000. Trend is up; volatility appears high; focus area is high variance.
- fail: value range 0.000 to 1.000 (spread 1.000), middle 50% spans 1.000. Trend is stable; volatility appears high; focus area is high variance.

Key Findings

- footfall: values range from 0.00 to 7300.00, average around 306.38. Trend appears down; review note: 18 anomalies, high variance.
- tempMode: values range from 0.00 to 7.00, average around 3.73. Trend appears up; review note: high variance.
- AQ: values range from 1.00 to 7.00, average around 4.33. Trend appears down; review note: high variance.
- USS: values range from 1.00 to 7.00, average around 2.94. Trend appears up; review note: high variance.
- CS: values range from 1.00 to 7.00, average around 5.39. Trend appears down; review note: 13 anomalies, high variance.
- VOC: values range from 0.00 to 6.00, average around 2.84. Trend appears up; review note: high variance.

AI-Powered Insights

Executive Summary: This report explains the analysis results for data.csv in clear language. The findings are based on the same data used in the website dashboard and do not introduce different calculations.

Data Overview: The dataset includes approximately 944 records and 10 variables. The analysis focused on trend behavior, consistency, unusual values, and confidence indicators.

Key Findings: Main focus areas include footfall:18 anomalies, high variance, tempMode:high variance, AQ:high variance, USS:high variance, CS:13 anomalies, high variance, VOC:high variance, RP:high variance, IP:high variance, Temperature:high variance, fail:high variance. Trend signals show footfall:down, tempMode:up, AQ:down, USS:up, CS:down, VOC:up, RP:up, IP:down, Temperature:up, fail:stable.

AI-Powered Insights: The observed changes may reflect shifts in operating conditions, collection methods, or natural process variation. Results should be reviewed with context from the experiment timeline.

Confidence & Reliability: The analysis is generally reliable for decision support, but confidence can be reduced when unusual values or missing entries are present.

Technical Review (Advanced)

Technical Review (Advanced):

- footfall: mean=306.3814, std=1082.6067, min=0.0000, max=7300.0000, IQR=109.0000, CV=3.5335, trend=down, focus=18 anomalies, high variance, confidence=0.98.
- tempMode: mean=3.7278, std=2.6772, min=0.0000, max=7.0000, IQR=6.0000, CV=0.7182, trend=up, focus=high variance, confidence=1.0.
- AQ: mean=4.3252, std=1.4384, min=1.0000, max=7.0000, IQR=3.0000, CV=0.3326, trend=down, focus=high variance, confidence=1.0.
- USS: mean=2.9396, std=1.3837, min=1.0000, max=7.0000, IQR=2.0000, CV=0.4707, trend=up, focus=high variance, confidence=1.0.
- CS: mean=5.3941, std=1.2693, min=1.0000, max=7.0000, IQR=1.0000, CV=0.2353, trend=down, focus=13 anomalies, high variance, confidence=0.99.
- VOC: mean=2.8422, std=2.2733, min=0.0000, max=6.0000, IQR=4.0000, CV=0.7999, trend=up, focus=high variance, confidence=1.0.
- RP: mean=47.0434, std=16.4231, min=19.0000, max=91.0000, IQR=24.0000, CV=0.3491, trend=up, focus=high variance, confidence=1.0.
- IP: mean=4.5657, std=1.5993, min=1.0000, max=7.0000, IQR=3.0000, CV=0.3503, trend=down, focus=high variance, confidence=1.0.
- Temperature: mean=16.3316, std=5.9748, min=1.0000, max=24.0000, IQR=7.0000, CV=0.3658, trend=up, focus=high variance, confidence=1.0.
- fail: mean=0.4163, std=0.4932, min=0.0000, max=1.0000, IQR=1.0000, CV=1.1847, trend=stable, focus=high variance, confidence=1.0.

Technical Recommendations (Advanced)

Technical Recommendations (Advanced):

- footfall: high coefficient of variation (CV=3.5335); apply robust thresholding and recalibration checks.
- footfall: anomaly-heavy profile detected; run outlier root-cause analysis against raw capture logs.
- footfall: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- tempMode: high coefficient of variation (CV=0.7182); apply robust thresholding and recalibration checks.
- tempMode: persistent upward trend; evaluate drift alarms and upper tolerance boundaries.

- AQ: high coefficient of variation (CV=0.3326); apply robust thresholding and recalibration checks.
- AQ: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- USS: high coefficient of variation (CV=0.4707); apply robust thresholding and recalibration checks.
- USS: persistent upward trend; evaluate drift alarms and upper tolerance boundaries.
- CS: anomaly-heavy profile detected; run outlier root-cause analysis against raw capture logs.
- CS: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- VOC: high coefficient of variation (CV=0.7999); apply robust thresholding and recalibration checks.

Special AI Recommendations

Priority actions below are generated from trend volatility, confidence, and focus-area signals.

Abstract: The analysis processed 944 rows across 10 columns to evaluate quality, trend behavior, and operational risk.

Introduction: Recommendations are mapped to Performance (25%), Innovation (25%), Operational Impact (25%), and Responsible AI (25%).

Materials and Methods: Data quality, distribution spread, quartiles, trend direction, focus-area tags, and confidence values were evaluated per metric.

Results:

- footfall: mean=306.3814, std=1082.6067, range=0.0000..7300.0000, IQR=109.0000, CV=3.534, trend=down, focus=18 anomalies, high variance, confidence=0.98.
- tempMode: mean=3.7278, std=2.6772, range=0.0000..7.0000, IQR=6.0000, CV=0.718, trend=up, focus=high variance, confidence=1.0.
- AQ: mean=4.3252, std=1.4384, range=1.0000..7.0000, IQR=3.0000, CV=0.333, trend=down, focus=high variance, confidence=1.0.
- USS: mean=2.9396, std=1.3837, range=1.0000..7.0000, IQR=2.0000, CV=0.471, trend=up, focus=high variance, confidence=1.0.
- CS: mean=5.3941, std=1.2693, range=1.0000..7.0000, IQR=1.0000, CV=0.235, trend=down, focus=13 anomalies, high variance, confidence=0.99.
- VOC: mean=2.8422, std=2.2733, range=0.0000..6.0000, IQR=4.0000, CV=0.800, trend=up, focus=high variance, confidence=1.0.
- RP: mean=47.0434, std=16.4231, range=19.0000..91.0000, IQR=24.0000, CV=0.349, trend=up, focus=high variance, confidence=1.0.
- IP: mean=4.5657, std=1.5993, range=1.0000..7.0000, IQR=3.0000, CV=0.350, trend=down, focus=high variance, confidence=1.0.
- Temperature: mean=16.3316, std=5.9748, range=1.0000..24.0000, IQR=7.0000, CV=0.366, trend=up, focus=high variance, confidence=1.0.
- fail: mean=0.4163, std=0.4932, range=0.0000..1.0000, IQR=1.0000, CV=1.185, trend=stable, focus=high variance, confidence=1.0.

Discussion:

- Risk stratification: high risk=9, medium risk=1, low-confidence=0 metrics.
- High-priority metrics: footfall, tempMode, AQ, USS, CS, VOC, RP, IP, Temperature.
- Medium-priority metrics: fail.
- Recommendation confidence improves when calibration records, sampling context, and anomaly root-cause notes are attached to each run.

Conclusion: The dataset supports decision guidance when recommendations are executed with metric-level evidence, uncertainty disclosure, and safety checks.

Recommended Next Actions (criteria-aligned):

- 1) [Performance 25%] footfall: anomaly concentration detected; verify raw acquisition logs, sensor drift, and boundary thresholds.

- 2) [Innovation 25%] footfall: high variability (CV=3.534); add adaptive control limits and confidence-banded recommendations.
- 3) [Performance 25%] footfall: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.
- 4) [Innovation 25%] tempMode: high variability (CV=0.718); add adaptive control limits and confidence-banded recommendations.
- 5) [Performance 25%] tempMode: upward trend detected; review upper tolerance limits and likely drift drivers.
- 6) [Innovation 25%] AQ: high variability (CV=0.333); add adaptive control limits and confidence-banded recommendations.
- 7) [Performance 25%] AQ: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.
- 8) [Innovation 25%] USS: high variability (CV=0.471); add adaptive control limits and confidence-banded recommendations.
- 9) [Performance 25%] USS: upward trend detected; review upper tolerance limits and likely drift drivers.
- 10) [Performance 25%] CS: anomaly concentration detected; verify raw acquisition logs, sensor drift, and boundary thresholds.
- 11) [Operational Impact 25%] Log metric-level evidence, confidence tags, and safety outcomes in a consistent audit trail for transparent reporting.

Recommendations

Detailed Recommendations:

Priority 1 - Immediate Monitoring:

- footfall: focus=18 anomalies, high variance, trend=down, CV=3.534, confidence=0.98. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- tempMode: focus=high variance, trend=up, CV=0.718, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- AQ: focus=high variance, trend=down, CV=0.333, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- USS: focus=high variance, trend=up, CV=0.471, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- CS: focus=13 anomalies, high variance, trend=down, CV=0.235, confidence=0.99. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- VOC: focus=high variance, trend=up, CV=0.800, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.

Priority 2 - Investigation Actions:

- Compare unusual spikes and drops against experiment logs, equipment events, and environment changes from the same time window.
- Validate whether trend changes are expected process behavior or indicate drift needing correction.

Priority 3 - Improvement Actions:

- Standardize data collection checkpoints (calibration, timestamp validation, completeness checks) before each run.
- Add weekly review of confidence and focus-area flags to improve reliability over time.

Priority 4 - Governance and Safety:

- Require human approval for decisions when confidence is low or quality warnings are present.
- Keep a decision log linking each action to the supporting metric evidence and confidence value.

Priority Recommendation Guide



Priority Level	Meaning	Recommended Action
High Priority	Immediate attention required	Priority 1 - Immediate Monitoring:
Moderate Attention	Track closely and follow up	Priority 2 - Investigation Actions:
Low Concern	Routine monitoring is sufficient	Priority 3 - Improvement Actions:

Confidence & Reliability

The analysis is generally reliable when interpreted with the same context shown in the website dashboard. Attention is recommended for: footfall, tempMode, AQ, USS, CS, VOC, RP, IP, Temperature, fail due to unusual patterns or data quality concerns. Missing values or unusual spikes can reduce certainty, so regular monitoring and data quality checks are advised.

Document Notes

- Document Context:
- Source file: data.csv
 - File type: csv
 - Structural profile: rows=944, columns=10
 - This report is generated for a general lab-review workflow and operational decision support.

Legal & Compliance Context

- The report automatically adapts reference guidance to detected data context. Domain tags: general.
- Apply recommendations with human oversight and retain an auditable decision log linking actions to evidence.
 - Confirm data minimization, access control, and retention limits before operational use.

AI Responsibility & Transparency

This report is generated using AI technologies provided through Microsoft Azure. Recommendations are based on available data and are intended to support decision-making. Validate critical decisions with domain experts.

Appendix

Additional summary details are provided below for reference.

Metric	Mean	Std	Min	Max	Trend	Focus Area	Confidence
footfall	306.3814	1082.6067	0.0000	7300.0000	down	18 anomalies, high variance	0.98
tempMode	3.7278	2.6772	0.0000	7.0000	up	high variance	1.0
AQ	4.3252	1.4384	1.0000	7.0000	down	high variance	1.0
USS	2.9396	1.3837	1.0000	7.0000	up	high variance	1.0

Metric	Mean	Std	Min	Max	Trend	Focus Area	Confidence
CS	5.3941	1.2693	1.0000	7.0000	down	13 anomalies, high variance	0.99
VOC	2.8422	2.2733	0.0000	6.0000	up	high variance	1.0
RP	47.0434	16.4231	19.0000	91.0000	up	high variance	1.0
IP	4.5657	1.5993	1.0000	7.0000	down	high variance	1.0
Temperature	16.3316	5.9748	1.0000	24.0000	up	high variance	1.0
fail	0.4163	0.4932	0.0000	1.0000	stable	high variance	1.0

Detailed Data Table

Metric	Count	Mean	Std	Min	Max
footfall	944.0	306.3814	1082.6067	0.0	7300.0
tempMode	944.0	3.7278	2.6772	0.0	7.0
AQ	944.0	4.3252	1.4384	1.0	7.0
USS	944.0	2.9396	1.3837	1.0	7.0
CS	944.0	5.3941	1.2693	1.0	7.0
VOC	944.0	2.8422	2.2733	0.0	6.0
RP	944.0	47.0434	16.4231	19.0	91.0
IP	944.0	4.5657	1.5993	1.0	7.0
Temperature	944.0	16.3316	5.9748	1.0	24.0
fail	944.0	0.4163	0.4932	0.0	1.0

References (AI Use)

Expelexia Analytics Team. (2026). Automated dashboard analysis report for data.csv [Internal report]. Expelexia.

Microsoft. (n.d.). Azure OpenAI documentation. Microsoft Learn. Retrieved March 26, 2026, from <https://learn.microsoft.com/azure/ai-services/openai/>

Microsoft. (n.d.). Responsible AI resources. Microsoft Learn. Retrieved March 26, 2026, from <https://learn.microsoft.com/azure/ai-foundry/responsible-ai/>

National Institute of Standards and Technology. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST. Retrieved March 26, 2026, from <https://www.nist.gov/itl/ai-risk-management-framework>

Organisation for Economic Co-operation and Development. (2019). OECD principles on artificial intelligence. OECD AI Policy Observatory. Retrieved March 26, 2026, from <https://oecd.ai/en/ai-principles>

ReportLab Inc. (n.d.). ReportLab user guide. Retrieved March 26, 2026, from <https://www.reportlab.com/documentation/>